Queen's University Belfast

School of Mathematics and Physics

Modelling and Simulation of Traffic Flow

MTH3024: Modelling and Simulation

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Abstract

Traffic flow is simulated using the Intelligent Driver Model (IDM) on a circular road. The relationship between traffic density, mean speed and traffic flow is analysed in the macroscopic and microscopic regime and compared to the literature, allowing the critical density of the system to be identified.

A slip road is introduced that allows vehicles to enter and exit the motorway stochastically. We demonstrate that increasing the length of the slip road increases the density of vehicles on the ring road and reduces global traffic flow once the critical density is exceeded. Additionally, simulations with different speed limits show that traffic flow increases with speed up to a limiting value. We thus suppose ideal values for the length of a slip road and the speed limit.

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1 Introduction

Understanding traffic flow is essential for designing efficient transportation systems, reducing congestion, and improving road safety. Car following models allow us to study these effects. This project uses the Intelligent Driver Model (IDM), which, since its introduction, has become a foundational tool in traffic flow modelling. This model simulates reactions to the motion of the preceding vehicle, with each car assumed to follow identical behaviour rules.

This project aims to answer the following questions:

- How do the macroscopic and microscopic approaches to the model compare?
- How does the addition of a slip road affect the flow of the motorway and how long should it be?
- How does traffic flow change with an increase in the speed limit on the ring road?

The full simulation code and animation outputs used in this project are available in the public GitHub repository: `https://github.com/adam8higgins/Traffic-Flow-Model`

2 Methodology

2.1 Intelligent Driver Model

The IDM, which is described by Treiber and Kesting [2], adjusts the acceleration of each car according to the headway s_i and the speed difference (Δv_i) between vehicles. The cars are enumerated such that vehicle $i-1$ is directly ahead of vehicle i. The headway of car i, s_i, is defined as the distance between the front of car i and the rear of car $i-1$ (the car in front):

$$s_i = x_{i-1} - x_i - l, \tag{1}$$

where l is the length of each car, and $(\Delta v_i) = v_i - v_{i-1}$ is the difference in speed between car i and car $i-1$.

We define a "desired headway", s_i^*, which we aim to maintain by adjusting the acceleration of car i. Reasonably, we can express this as

$$s_i^* = s_{min} + \max\left(0, v_i T + \frac{v_i(\Delta v_i)}{2\sqrt{ab}}\right), \tag{2}$$

where:

- s_{min} is the minimum headway that *must* be maintained,
- T is the time gap,
- a and b are defined as the maximum acceleration and comfortable braking, respectively.

The other term, $\frac{v_i(\Delta v_i)}{2\sqrt{ab}}$, is used to simulate dynamical braking behaviour when approaching the car in front. The acceleration, $\dot{v}$, substituted into the equations of motion is then given as

$$\dot{v}_i = a\left[1 - \left(\frac{v_i}{v_{max}}\right)^{\delta} - \left(\frac{s_i^*}{s_i}\right)^2\right], \tag{3}$$

where:

- v_{max} is a constant speed limit (which may also be considered the desired speed),
- δ is a dimensionless value called the acceleration exponent.

Using this, we may simulate the positions of each car for a range of times separated by the time-step, Δt. $(\frac{s_i^*}{s_i})^2$ ensures that the vehicle decelerates when the headway falls below the threshold.
To prevent negative velocities, if $v_{\text{new}} < 0$ the time-step is rescaled so the vehicle stops exactly at zero:

$$(\Delta t)^* = \frac{-v}{\dot{v}}.$$

The positions and velocities of cars on the ring road must also be defined at the beginning of the simulation; for this, we suppose that the cars are in a queue and possess a linear spacing of the minimum separation, s_{min}.

2.2 Measuring System Parameters

To validate our IDM implementation, we must define measurable global parameters. We will use the average velocity of the cars given by:

$$\bar{v} = \frac{1}{N}\sum_{i=1}^{N} v_i \tag{4}$$

where N is the number of cars. The density ρ, is defined as:

$$\rho = N/L,$$

where L is the circumference of the ring road. The flow, Q, is the product of the average velocity of the cars and the density:

$$Q = \rho\bar{v},$$

The graphs Q vs. ρ, Q vs. $\bar{v}$, and $\bar{v}$ vs. ρ follow characteristic shapes reported by Hoogendoorn [3], allowing us to verify our IDM implementation. These quantities describe the global (macroscopic) behaviour of the system. A corresponding local (microscopic) regime is obtained by recording vehicle velocities and passage times with a virtual speed camera, which allows us to compare how closely local measurements match the global averages.

3 Results and Discussion

To construct Fig. [1], the model was simulated using the constants given in Table 1, varying the number of cars, N, until the ring road was at capacity. The simulation ran for 20 seconds without recording any parameters, so the transient effects that arise from the initial conditions were eliminated. Once completed, the simulation continued, allowing the global and local parameters described in sec. 2.2 to be evaluated.

Symbol	Description	Value Used
v_0	Desired speed	70 mph
a_{max}	Maximum acceleration	1.0 m/s^2
b	Comfortable deceleration	3.0 m/s^2
T	Time gap	1.5 s
δ	Acceleration exponent	4
s_{min}	Minimum gap	2.0 m
l	Vehicle length	4.5 m
L	Ring road circumference	1000 m
Δt	Time step	0.1 s

Table 1: Intelligent Driver Model parameters and simulation values used in the simulations.

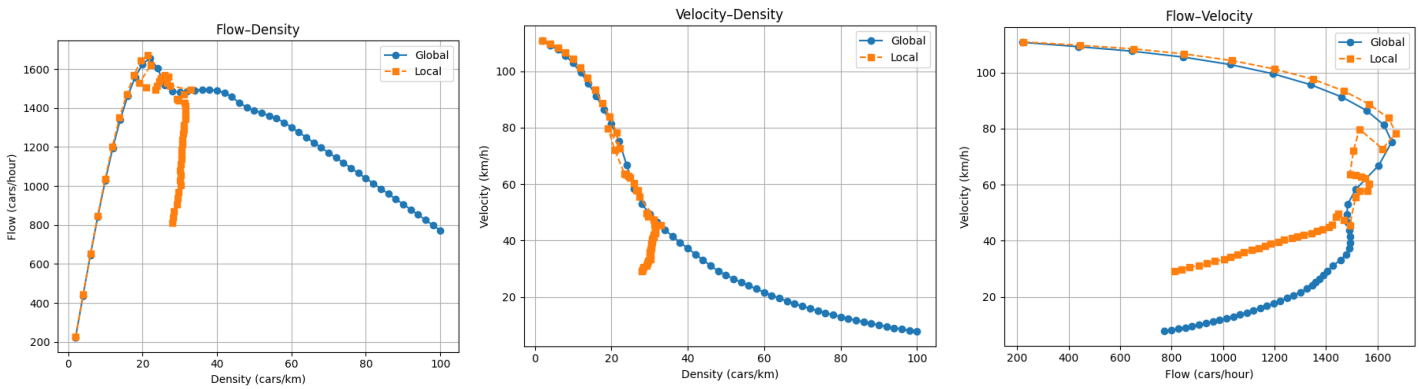

Figure 1: The three fundamental diagrams—flow–density, flow–velocity, and velocity–density—constructed using the parameters in 2.2. Each plot shows both the global (blue) and local (orange) regimes, allowing direct comparison between the macroscopic equilibrium behaviour of the system and the microscopic measurements obtained from the simulation.

Figure 1, reveals some key features of simulated traffic dynamics

1. **Model Validation**: For the global regime, Fig. [1] replicates the results obtained by Hoogendorn [3], verifying the validity of both our implementation of the IDM and the equations used to obtain equilibrium solutions.
2. **Low-density Equilibrium**: From all 3 graphs a key observation is that the local regime coincides with the global equilibrium curve at low densities. This tells us the global regime behaves as expected and provides a reliable benchmark to which alternative solutions can be compared to.
3. **Critical Density**: The global-local equivalence persists up to a critical density, ρ_c, beyond which the two regimes diverge significantly. For the values stated in Table 1, we find $\rho_c \approx 31$ cars km^{-1}. For $\rho > \rho_c$, the global regime exhibits a sharp reduction in both velocity and flow.

4 Extension: The addition of a slip road

To extend our traffic simulation, we introduced a slip road where vehicles can enter the main carriageway. The objective was to investigate how merging behaviour influences overall traffic performance. The slip road is constructed as a separate lane of finite length, with vehicles being introduced stochastically at a probability of P_{in} per time step.

At each time step, slip road vehicles merge only if the distances to both the car in front and behind on the main road exceed s_{min}, defined by the IDM. If this condition is not met at any point, the vehicle will queue at the end of the slip road where it must wait for a gap in traffic.

Since the road is circular, and vehicles flow in stochastically, an exit with removal probability P_{out} prevents the ring road from becoming full. Moreover, whenever a car merges, there is a change in local density that triggers deceleration, which is of particular interest to study. We therefore measure the global flow, Q (sec. 2.2) and assess how it varies with length, L_s, and speed limit.

4.1 How the length of the slip road affects traffic flow.

Figure 2 was constructed by measuring the flow, Q, while varying the length of the slip road, L_s. The speed limit was taken as 70 mph, with probabilities $P_{in} = P_{out} = 0.2$.

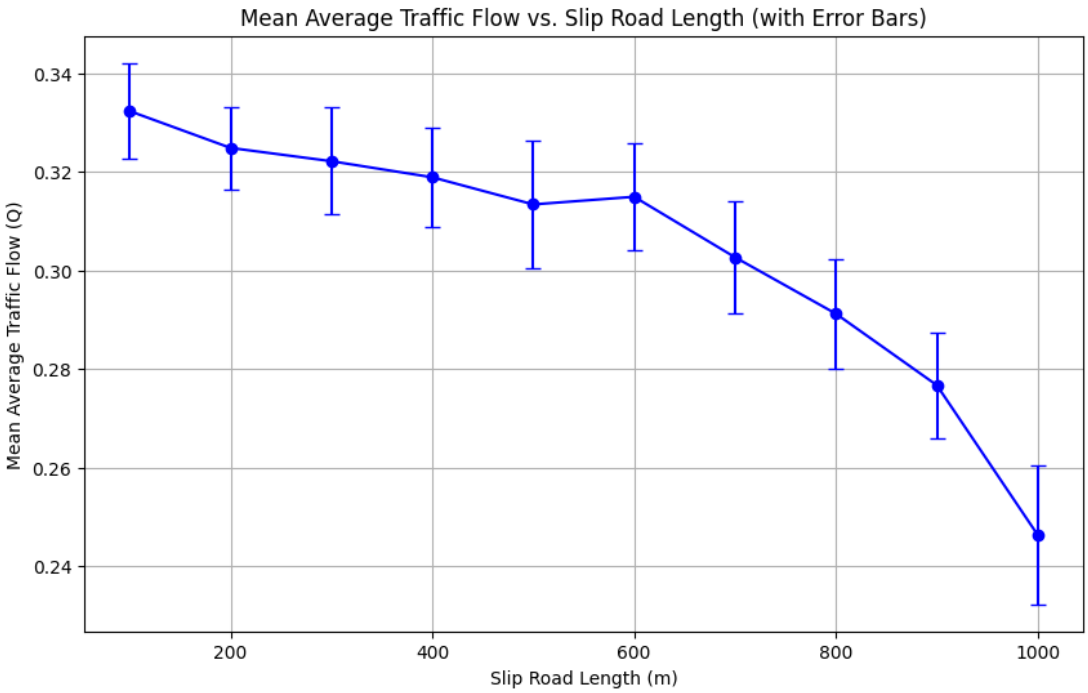

Figure 2: Mean traffic flow on the ring road, Q, as a function of slip road length, L_s. The mean flow was evaluated by running the simulation 30 times, with the error bars being the standard deviation.

Fig. 2 shows clearly that the flow on the ring road decreases as the length of the slip road increases. This is expected as a longer slip road allows merging vehicles more time to accelerate toward the main road equilibrium velocity, thereby increasing the likelihood that a car successfully merges. Thus, the operating density, ρ, of the ring road decreases as L_s increases.

From the fundamental diagrams in sec. 3,the flow is maximised at the critical density, ρ_c, and is substantially decreased for $\rho > \rho_c$. Thus, L_s effectively controls whether the system remains in free flow or enters congestion. Increasing the slip road length leads to reduced macroscopic flow despite smoother individual merges, demonstrating that improved merging efficiency does not necessarily increase capacity. For $L_s \approx 600$ metres, the flow drops significantly, suggesting that the operating density has exceeded the critical density ρ_c for these parameters.

4.2 Effect of speed limit in the presence of a slip road

With $L_s = 800m$, and P_{in} and P_{out} being the same as in 4.1, the figures 3 and 4 were produced by varying the IDM's speed limit parameter, $v_{\max}$.

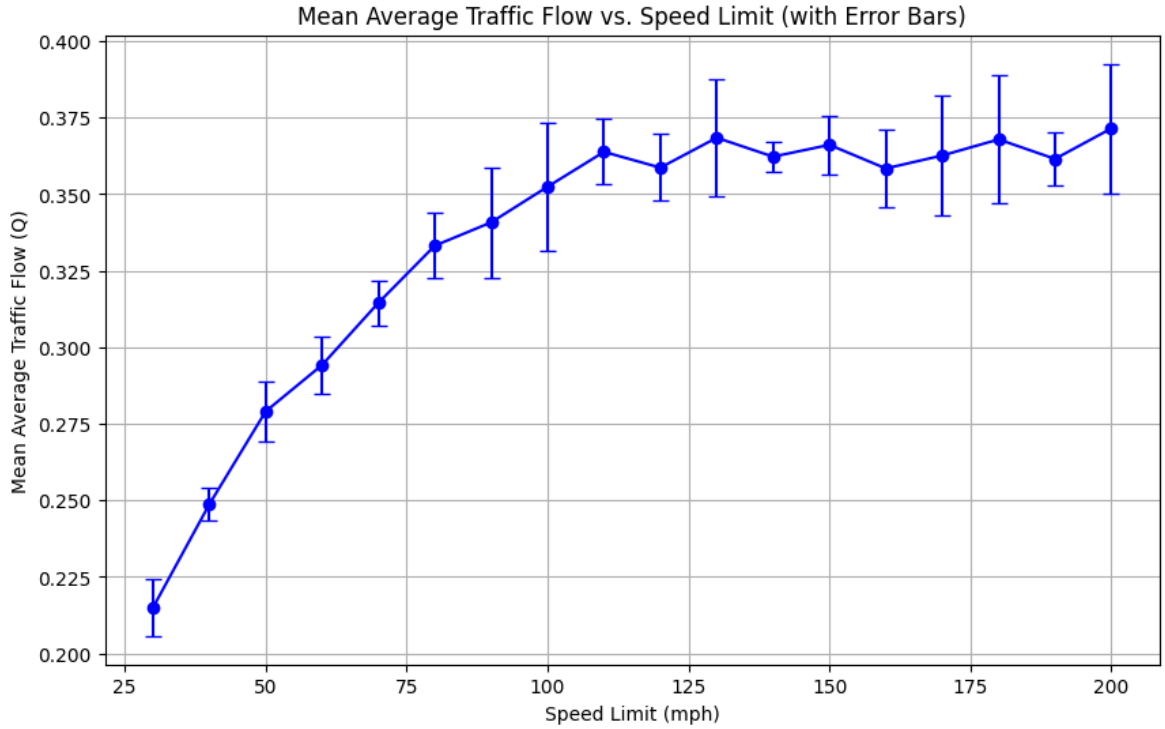

Figure 3: Mean traffic flow, Q, as a function of speed limit, v_{max}, calculated by running the simulation 30 times.

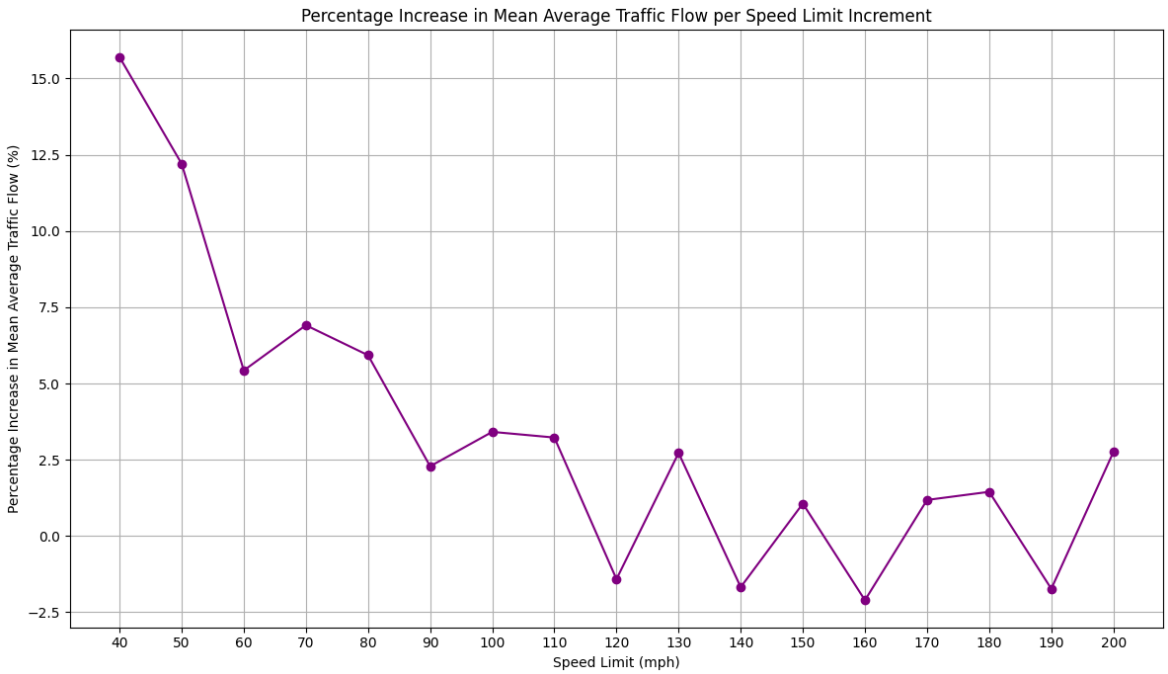

Figure 4: Percentage increase in the mean traffic flow, Q, against the speed limit, v_{max}, from the 30 simulations.

Figure 4 reframes the data given in fig. 3, to show clearly the limits of increasing v_{max}. For low speed limits, the equilibrium velocity is lower, limiting the magnitude of the flow. Increasing $v_{\max}$ raises the free flow speed and therefore increases Q. We observe this in fig. 4, where between 40 and 50 mph there is a 12.2% increase in traffic flow. However, at higher speed limits, smaller time headways at higher velocities make deceleration waves more pronounced. As a result, further increases in speed limits cause smaller improvements in flow, e.g., between 80 and 90 mph traffic flow only increases by 2.3%.

From this we conclude that a higher speed limit on motorways is desired to manage flow. However, there is no significant increase in flow when the speed limit exceeds 80 mph. The model therefore predicts the existence of an effective optimal speed limit in the presence of inflow, consistent with known motorway designs. [5]

5 Conclusions, Limitations and Further Work

In this project, we used the Intelligent Driver Model to simulate traffic flow on a ring road and to study how slip roads and speed limits influence motorway performance. The questions posed in sec. 1 have now been addressed:

1. How do the macroscopic and microscopic approaches to the model compare?
   - The macroscopic regime may be used as a benchmark for system behaviour, since in sec. 3 we showed that the IDM exhibits a critical density ρ_c at which flow is maximised. For $\rho < \rho_c$, the local and global measurements agree, but once the density exceeds this threshold the relationship breaks down.
2. How does the addition of a slip road affect the flow of the motorway and how long should it be?
   - We recommend limiting slip-road length, as longer slip roads—while helping vehicles merge smoothly—also increases the density on the ring road. Once the density surpasses the critical density ρ_c, the system transitions from free flow to congestion. Under the conditions in 4.1, slip roads longer than around 600 metres reduce overall flow.
3. How does traffic flow change with an increase in the speed limit on the ring road?
   - We recommend setting the speed limit within $v_{max} \approx 70$–80 mph, since increasing the limit increases traffic flow only up to this range. Beyond this, the flow does not improve significantly as higher speeds increase both average velocity and the severity of required deceleration, promoting congestion.

However, the IDM has several limitations as a car-following model. It does not account for human factors such as reaction times, and collisions are prevented entirely, which is unrealistic in the high-speed, harsh-braking scenarios used in figure 4. Merging behaviour can also trigger large, abrupt decelerations on the ring road, potentially obscuring some results. Future work will therefore focus on adopting a more realistic acceleration law than eq. (3) and incorporating more varied, stochastic driver behaviours to better reflect real motorway conditions. We would also implement randomly generated initial conditions, instead of the "queue of traffic" that was supposed in sec. 2.1, to produce a more realistic simulation.

Acknowledgements

The authors acknowledge the use of generative AI tools in generating portions of the Python code used to create the animation outputs available in the project GitHub repository. These animations were used to assist with visualising the behaviour of the simulation and validating the implementation of the model.

References

[1] R. E. Chandler, R. Herman, and E. W. Montroll. Traffic dynamics: Studies in car following. *Operations Research*, 6(2):165–184, 1958.

[2] M. Treiber and A. Kesting. *Traffic Flow Dynamics*. Springer, Berlin, 2013.

[3] S. P. Hoogendoorn. *Traffic Flow Theory and Simulation*, 2016, p. 82. Available at: `https://ocw.tudelft.nl/wp-content/uploads/Chapter-4.-Fundamental-diagrams.pdf`

[4] A. Higgins. Traffic Flow Simulation Model (IDM). Available at: `https://github.com/adam8higgins/Traffic-Flow-Model`

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A Appendix:

A.1 Appendix A: Additional Results for the Slip Road Length Extension

In Section 4.1 traffic flow on the ring road decreases as the length of the slip road increases. To obtain the flow values presented in the main results, the mean vehicle density and mean vehicle velocity were recorded during each simulation run.

Figure 5 presents the simulation outputs used to compute the flow values.

The left graph shows the mean average density as a function of slip road length. A gradual decrease in density can be observed as the slip road length increases. This occurs because longer slip roads allow vehicles more time to merge safely onto the main carriageway, reducing the likelihood of vehicles accumulating on the ring road.

The right graph shows the mean average velocity of vehicles on the ring road. The velocity remains relatively stable for shorter slip road lengths but begins to decrease slightly for longer slip roads. This reflects the increased disturbance to the traffic stream caused by vehicles merging over a larger region of the road.

Together, these two quantities determine the traffic flow reported in Section 4.1. Alike to the main results, the error bars represent the variability across the 30 simulation runs and illustrate the stochastic nature of vehicle arrivals and merging behaviour in the model.

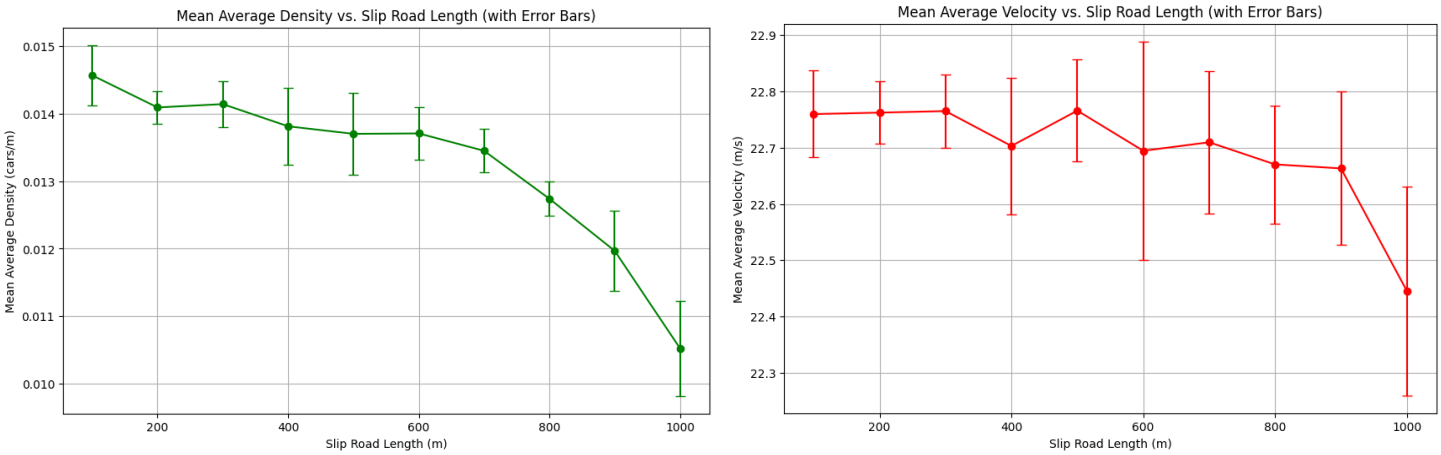

Figure 5: Additional simulation outputs for the slip road extension model. Left: mean vehicle density as a function of slip road length. Right: mean vehicle velocity as a function of slip road length. These were used to compute the traffic flow results presented in the main report.

A.2 Appendix B: Additional Results for the Speed Limit Experiment

In Section 4.2, the effect of increasing the speed limit on the ring road was investigated in the presence of a slip road with length $L_s = 800m$. To obtain the flow values presented in Figures 3 and 4, the mean vehicle density and mean vehicle velocity were recorded during each simulation.

Figure 6 shows the simulation outputs used to compute the flow values.

The left graph shows the mean average density as a function of the speed limit. A clear decrease in density can be observed as the speed limit increases from lower values. This occurs because vehicles travel faster and therefore spend less time occupying the ring road, reducing the average density of cars within the system.

The right graph shows the corresponding mean average velocity of vehicles on the ring road. As expected, the mean velocity increases rapidly for lower speed limits before gradually levelling off as the speed limit becomes large. This behaviour reflects how vehicles interact and the headway constraints in the IDM, which limit how fast cars can realistically travel even when the imposed speed limit is high.

Together, these two quantities determine the traffic flow reported in Section 4.2. As in the main results, the error bars represent the variability across the 30 simulation runs and illustrate the stochastic nature of the inflow and merging behaviour in the model.

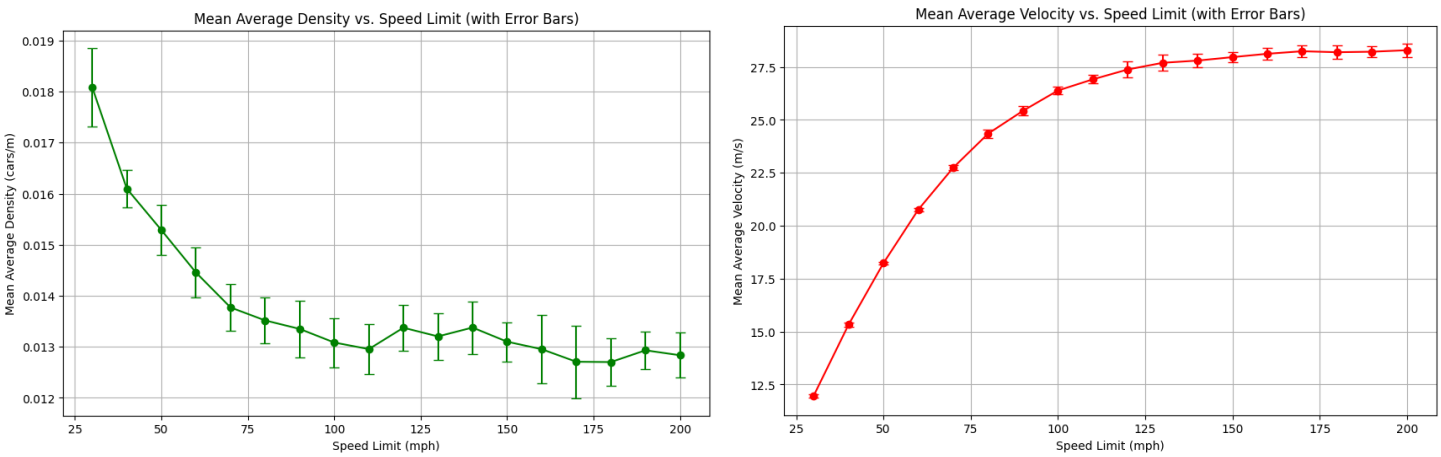

Figure 6: Additional simulation outputs for the speed limit experiment. Left: mean vehicle density as a function of speed limit. Right: mean vehicle velocity as a function of speed limit. These were used to compute the traffic flow results presented in the main report.